GLUING MORPHISMS ALONG A PROPER MODIFICATION IS CONTROLLED BY THE FORMAL COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a uniqueness step via the gluing lemma, an etale-descent reduction to the affine Noetherian case, an equality over the self fibre product, and a coequalizer-system construction finished by essential constancy, all inside the delivered span

Editorial note. The label proposition-glue-modification names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- The formal completion X_{/T} of an algebraic space along a closed subset, and its functoriality under the modification f : X' to X which is an isomorphism over the complementary open
- The faithful gluing lemma of the same chapter (equality of morphisms from agreement on the open and after completion)
- The coequalizer construction in the pushouts section of Pushouts of Spaces, including essential constancy of the system X_n with value X

Source reference. Original reference 0BLN: sources/stacks-restricted/pinned-original-source.tex, source lines 6484–6484. Prerequisite pointer invoked at proof line(s) 6657; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGG: sources/stacks-chow/tags.txt, source lines 13194–13194. Prerequisite pointer invoked at proof line(s) 6698; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGK: sources/stacks-chow/tags.txt, source lines 13198–13198. Prerequisite pointer invoked at proof line(s) 6713; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0AGL: sources/stacks-chow/tags.txt, source lines 13199–13199. Prerequisite pointer invoked at proof line(s) 6714; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 0AGF: sources/stacks-chow/tags.txt, source lines 13193–13193. Prerequisite pointer invoked at proof line(s) 6671; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let S be a scheme. Let X be a locally Noetherian algebraic space over S. Let $T \subset |X|$ be a closed subset with complementary open subspace $U \subset X$. Let $f : X' \to X$ be a proper morphism of algebraic spaces such that $f^{-1}(U) \to U$ is an isomorphism. For any algebraic space W over S the map*

$$\mathrm{Mor}_S(X, W) \longrightarrow \mathrm{Mor}_S(X', W) \times_{\mathrm{Mor}_S(X'_{/T}, W)} \mathrm{Mor}_S(X_{/T}, W)$$

is bijective.

Proof. Let $w' : X' \to W$ and $\hat{w} : X_{/T} \to W$ be morphisms which determine the same morphism $X'_{/T} \to W$ by composition with $X'_{/T} \to X$ and $X'_{/T} \to X_{/T}$. We have to prove there exists a unique morphism $w : X \to W$ whose composition with $X' \to X$ and $X_{/T} \to X$ recovers w' and $\hat{w}$. The uniqueness is immediate from Lemma 0BLN.

The assumptions on T and f are preserved by base change by any étale morphism $X_1 \to X$ of algebraic spaces. Since formal algebraic spaces are sheaves for the étale topology and since we already have the uniqueness, it suffices to prove existence after replacing X by the members of an étale covering. Thus we may assume X is an affine Noetherian scheme.

Assume X is an affine Noetherian scheme. We will construct the morphism $w : X \to W$ using the material in Pushouts of Spaces, Section 0AGF. It makes sense to read a little bit of the material in that section before continuing the read the proof.

Set $X'' = X' \times_X X'$ and consider the two morphisms $a = w' \circ \mathrm{pr}_1 : X'' \to W$ and $b = w' \circ \mathrm{pr}_2 : X'' \to W$. Then we see that a and b agree over the open U and that $a_{/T}$ and $b_{a/T}$ agree (as these are both equal to the composition $X''_{/T} \to X_{/T} \to W$ where the second arrow is $\hat{w}$). Thus by Lemma 0BLN we see $a = b$.

Denote $Z \subset X$ the reduced induced closed subscheme structure on T. For $n \geq 1$ denote $Z_n \subset X$ the nth infinitesimal neighbourhood of Z. Denote $w_n = \hat{w}|_{Z_n} : Z_n \to W$ so that we have $\hat{w} = \mathrm{colim}\, w_n$ on $X_{/T} = \mathrm{colim}\, Z_n$. Set $Y_n = X' \amalg Z_n$. Consider the two projections

$$s_n, t_n : R_n = Y_n \times_X Y_n \longrightarrow Y_n$$

Let $Y_n \to X_n \to X$ be the coequalizer of s_n and t_n as in Pushouts of Spaces, Section 0AGF (in particular this coequalizer exists, has good properties, etc, see Pushouts of Spaces, Lemma 0AGG). By the result $a = b$ of the previous parapgraph and the agreement of w' and $\hat{w}$ over $X'_{/T}$ we see that the morphism

$$w' \amalg w_n : Y_n \longrightarrow W$$

equalizes the morphisms s_n and t_n. Hence we see that for all $n \geq 1$ there is a morphism $w^n : X_n \to W$ compatible with w' and w_n. Moreover, for $m \geq 1$ the composition

$$X_n \to X_{n+m} \xrightarrow{w^{n+m}} W$$

is equal to w^n by construction (as the corresponding statement holds for $w' \amalg w_{n+m}$ and $w' \amalg w_n$). By Pushouts of Spaces, Lemma 0AGK and Remark 0AGL the system of algebraic spaces X_n is essentially constant with value X and we conclude. $\square$